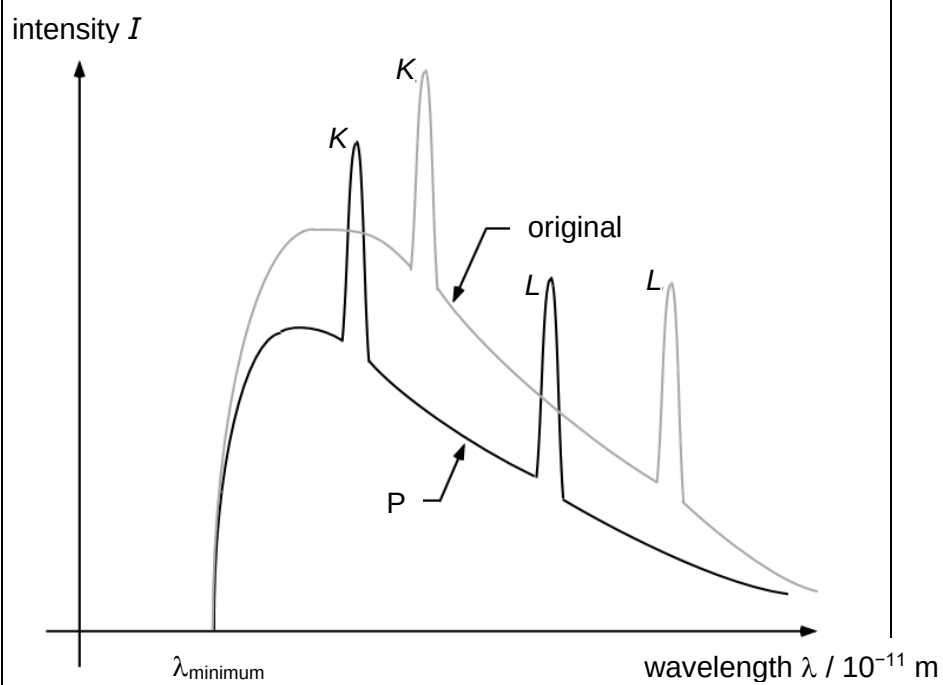


			[A1] for correct label of ε_0 and T/4, 3T/4, 5T/4, 7T/4 (no values required).	
6	(a)	(i)	1. energy = $0 - (-13.6) = 13.6 \text{ eV} = 13.6 \times 1.6 \times 10^{-19}$ $= 2.18 \times 10^{-18} \text{ J}$ B1	
			2. The electron of the atom will <u>absorb 10.2 eV of energy</u> and be <u>excited to $n = 2$ level</u> . (The incident electron will have 0.80 eV of energy left.) B1	
			3. Nothing happens (No change). The <u>energy of the photon does not match any transition</u> (from ground state). B1	
		(ii)	$\Delta E = -0.849 - (-13.6)$ $= 12.75 \text{ eV}$ $= 2.04 \times 10^{-18} \text{ J}$ C1 $\Delta E = \frac{hc}{\lambda}$ $\lambda = \frac{6.63 \times 10^{-34} (3.00 \times 10^8)}{2.04 \times 10^{-18}}$ $= 9.75 \times 10^{-8} \text{ m}$ or 97.5 nm M1 <u>Ultraviolet</u> radiation will be emitted. A1 (Do not accept UV.)	
	(b)	(i)	$\frac{hc}{\lambda_{\min}} = E_{k, \text{ initial}} = e(\Delta V)$ $E_k = 1.60 \times 10^{-19} (1.05 \times 1000) = 1.68 \times 10^{-14} \text{ J}$ C1 $\lambda_{\min} = \frac{6.63 \times 10^{-34} (3.00 \times 10^8)}{1.68 \times 10^{-14}}$ M1 $= 1.18 \times 10^{-11} \text{ m}$ A1	

(ii)



B1 Same λ_{minimum} , and lower intensity throughout.

B1 Both peaks shift left

7	(a)	(i)	progressive means there is energy transfer in the direction of	B1
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